BRICK WORK · MILESTONE 4.0 · STATUS REPORT

Prod runs on schema 4.0

Goal met on 28 Oct 2026 at 13:00. Every item has a Brick Work id, the catalogue comes from Rebrickable’s dump with no key, the collection was migrated from 3.3 with a proof, and Toni and lego-add use API v2.

6 days
23–28 Oct, planned 18–22 weeks

93
commits on main

3,661
tests, quarantine 0

11 min
prod down, 12:49–13:00

Gate #116, passed 28 Oct · released as 0.6.0 on 29 Oct · milestone closed 4 Nov · ADRs 0016, 0017, 0018

DONE

All twelve pieces closed

Item	Piece	Closed
M.1	Schema and connection	23 Oct
M.2	Catalogue builder	24 Oct
M.3	Shipped mappings	23 Oct
M.4	Figure mapping test	23 Oct
M.5	Acquired part lists	25 Oct
M.6	Copies and switching	25 Oct
M.7	**Migration with proof**	26 Oct
M.8	Core modules on item_id	27 Oct
M.9	Test-suite rewrite	27 Oct
M.10	Screens, 36 frames approved	28 Oct
M.11	**API v2, MCP, both clients**	28 Oct
M.12	**Onboarding, checks, cutover**	28 Oct

Achieved highlights: a migration proof that passed identically on the Mac, on test and on prod; 188 API v2 operations with an MCP that takes numbers; a fresh install that reaches a checkable set with no key.

AGAINST THE PLAN

What changed, and why

Change	Why
6 days, not 18–22 weeks	One piece at a time: gate, Codex review, real-data check
Toni App Lego tab added	About 45 v1 calls, missing from #113, found in M.11f
Catalogue entry gains theme	Migrated items lack properties until jobs re-fetch them
Runbook in rasp-infra	It is host glue, not app documentation
No BrickLink lists in CAT	The configuration the rehearsal proved; they load later
Pi first-import timing	Not measured; filed as #126

DECISIONS

Three decision records

ADR 0016

Data model 4.0

Brick Work ids for every item, a catalogue from Rebrickable’s dump, providers linked with a confidence, one migration with a proof.

ADR 0017

API v2 and its MCP

v1’s resources with the 4.0 identity. Number lookups default to BrickLink; ambiguous is 409; merged is 307. MCP tools take numbers, never ids.

ADR 0018

The cutover

One pinned catalogue, a rehearsal from a prod backup, the clients in the window, rollback by rename and image tag.

THE GATE · #116

Each check, with its evidence

Check	Evidence
M.1–M.12 closed	#103–#114 closed, the last three on 28 Oct
Proof PASS, rehearsed on test	#114: lego-test from the prod backup, pinned catalogue
Prod on 4.0, 3.3 file kept	#115: prod-1c23570; data/brickwork-3.3.db, 196 copies
Toni and lego-add on v2	House API on v2, lego-add lookup and an Undo write on prod
Catalogue in Borg, checked	brick-work-2026-10-28_13:03: catalogue.db, dump ok
Close-out	This report; PDF in docs/milestones/4.0-data-model.pdf

RETROSPECTIVE

What worked, what didn’t, what changes

Went well

Real-data checks found what tests missed: a price stored under the wrong colour, and 16% attached without the mapping layer.

The rehearsal found five runbook defects; prod’s numbers then matched it exactly.

Rollback rehearsed: 9 s to healthy.

Went badly

The client list was incomplete: the Lego tab came late, and an env template kept /api/v1 (410 for a minute).

GitHub comments went out as Ulf, not the bot.

The cutover-day Borg run pruned the 02:01 archive.

Improve

Applied: the runbook checks deployed client config and names the Borg prune.

Sent to the wiki-agent: a client inventory before a breaking API, a bot-only GitHub hook, a venv for worktree gates.

CARRIES OVER

Into milestone A and the backlog

Item	Where	State, 4 Nov
Re-plan A against 4.0	Before A starts (A's deck, #83, #88)	**Done** · 3 Nov; A closed, 0.8.0
Properties, pictures, prices	Re-fetched by jobs; Brickset about 4 days	None waiting on prod; 4 failed
30 manual identity decisions	Redo from migrate4-report/manual-decisions.csv	Open
Site API page still v1	#125	**Done** · 28 Oct
Pi first-import timing	#126	**Done** · 51 s, 16bf67e
Adding a set in the UI	#121, works on 4.0 on test; Ulf to confirm on prod	**Done** · 29 Oct
Retire the old BrickTracker	Ulf, no earlier than 17 Nov	Open

BRICK WORK · MILESTONE 4.0

Data model 4.0

Goal: Brick Work runs on schema 4.0 in prod. Every set, part, minifigure and colour has a Brick Work id; the catalogue comes from Rebrickable’s dump with no key; the collection is migrated from 3.3 with a proof; API v2 replaces v1.

Rough length: 18–22 weeks · Before milestone A · Gate: issue #116 · ADR 0016

WHY

BrickLink numbers cannot carry a catalogue that comes from Rebrickable

Today, 0.5

BrickLink’s type and number is the identity of everything. What cannot be translated becomes a “?” item. Sticker sheets, legs whose BrickLink id depends on colour, and assemblies have no place.

After 4.0

Brick Work gives every item its own id. BrickLink, Rebrickable and Brickset link to it with a confidence: matched, inferred, manual or to check, scoped to a colour where needed.

Spike #102 settled the source: Rebrickable’s dump is free to use with acknowledgement, BrickStore’s archive of BrickLink data has no shown permission. The identity touches 54 key references in 40 tables, so the whole schema is re-derived.

WHAT CHANGES FOR THE USER

A reference book and a notebook

Area	In 4.0
Two files	catalogue.db is rebuilt from the dump; brickwork.db holds the collection and is never touched by a rebuild
Adding a set	Offline and instant from the catalogue; a part-list picker when there is a choice
Part lists	Rebrickable by default; BrickLink’s list per set or by a global switch; “version n of m”
Keys	None needed to start; BrickLink adds prices, order import and its part lists
Migration	Collection, prices, keys and tokens, preferences; history stays in the old file
API	v2 by Brick Work id with number lookups; v1 answers 410; Toni and lego-add switch

THE WORK

Twelve pieces, about 20 weeks

Item	Piece	Weeks
M.1	Schema and connection	1
M.2	Catalogue builder	2.5
M.3	Shipped mappings	1
M.4	Figure mapping test	0.75
M.5	Acquired part lists	0.75
M.6	Copies and switching	1

Item	Piece	Weeks
M.7	Migration with proof	2
M.8	Core modules on item_id	4.5
M.9	Test-suite rewrite	2.5
M.10	Screens, frames first	2
M.11	API v2 and clients	1.5
M.12	Onboarding, checks, cutover	0.75

Plus 1 week of contingency. M.1, M.3 and M.4 can start together. The least certain lines are the core modules, the test rewrite and BrickStore’s file format.

THE GATE · #116

Prod runs on 4.0, proven against 3.3

- M.1–M.12 closed
- Migration proof PASS on a copy of prod on test, rehearsed with the cutover catalogue
- Cutover decided (#115), prod on 4.0, the 3.3 file kept beside it
- Toni and lego-add working against API v2
- Catalogue file in Borg, first archive checked
- Close-out: this deck becomes the milestone’s status report